Cut-regime data spec — definitions & collection

Companion to `docs/cut-regime-models.md` (equations and trust tiers). This file defines **what we record and what each label means**, so that every run is comparable and the maps can be built from it. Data lives in `data/cut-regime/runs.csv`.

Status: **v0.4** (2026-11-09). Thresholds marked `TBD` are set from a calibration set, not guessed. Change a definition only by editing this file and noting the change in §12.

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1. Scope and fixed series

One **series** = one fixed combination of:

- optics (lens focal length, `optics_id`), nozzle (`nozzle_id`, diameter `d_n`), gas grade and purity,
- material grade and nominal thickness `t_nom`,
- feature type (`web` / `flange` / `wall`), and face (see §5),
- focus offset convention (§3.3).

Changing any of these starts a new series. Do not pool series in the same map.

Data rows are **one sample per line** (one straight cut sample, §4.1). Corners and curves are separate feature classes (§7, §8) and are never averaged into straight-line data.

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2. Units and conventions

Quantity	Unit	Note
Power `P_W`	W	Measured/calibrated watts at the workpiece. Log `P_set_pct` as well.
Speed `v_m_min`	m/min	Shop unit. Convert to mm/s only in formulas.
Thickness `t`	mm	**Measured** with calliper at the cut, not catalogue. `t_nom` is the catalogue value.
Gas pressure `gas_p_bar`	bar	At the head. **State gauge (barg) or absolute (bar abs)** in the series header. Shop values are unstated; see models §5.
Focus `focus_f_mm`	mm	Physical: **negative = below the top surface** (shop practice: inside or past the material), positive = above. The controller/CAM value is recorded separately in `cam_focus_mm`.
Standoff `standoff_h_mm`	mm	Nozzle tip to surface.
Energy S_{beam}	J/mm²	$S_{\mathrm{beam}} = 0.06\,\dfrac{P_W}{v_{\mathrm{m/min}}\,t_{\mathrm{mm}}}$. **Beam-only** — default S.
Energy S_{eff}	J/mm²	Only if oxidation energy is estimated (models §2: $P_{\mathrm{eff}} = \eta_{\mathrm{abs}} P + P_{\mathrm{ox}}$). Leave blank otherwise.

Rule: every table, plot and comparison states whether S is `S_beam` or `S_eff`. Default is `S_beam`.

Sign and direction: travel direction is always the direction of the beam path on the machine (see §7). Record it, do not infer it later.

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3. Settings that must be recorded

3.1 Per run (in `runs.csv`)

`P_set_pct`, `P_W`, `v_m_min`, `focus_f_mm`, `gas_p_bar`, `standoff_h_mm`, `gas_purity_pct`.

3.2 Per series (in the series header, §10)

`optics_id`, `nozzle_id`, `nozzle_dn_mm`, `gas_grade`, `material_grade`, `t_nom_mm`, `section`, `feature`, machine id, date, operator.

3.3 Focus convention

Focus is measured from the **top surface** of the workpiece at the cut location. Physical sign: **negative = focus below the top surface** (shop practice: inside or past the material), positive = above. The controller/CAM value is logged separately in `cam_focus_mm`. If the height sensor reads from the web or a recessed channel, note it: it changes the effective focus (see `cut-regime-models.md` §8).

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4. Samples and windows

4.1 Straight-cut sample

- A straight segment of at least **40 mm** long, steady-state.
- **Sample window:** the middle **20 mm**, excluding the first and last **10 mm** (run-in and run-out). Record `cut_length_mm` and `sample_window_mm`.
- The window must be on a single face and feature. No corner, no fillet, no hole in the window.

4.2 One face at a time

A UB/T/PFC web and flanges are separate samples. Each face (`kept` / `scrap`, §7) is graded separately.

4.3 Don't judge in the window alone

Record the whole cut (dross, bridging, separation) but grade the edge in the window.

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5. Cut outcome — `through_code` (one per sample, mutually exclusive)

Code	Name	Criterion (apply in order, take the first that matches)
N0	No penetration	No light passes through the kerf over the window.
N1	Through, bridged	Full-thickness kerf (light passes) but the piece is still joined by melt, slag or a bridge.
N2	Forced separation	Separates only with hand force, a tap, or a hammer.
N3	Free separation	Releases by gravity or minimal handling; **no tool, no hammer, no tearing**.

Decision rules:

- **"Cut"** in the PDF sense (penetration boundary, `S_min`) = **N1 or better**, continuous through-kerf.
- **"Separated"** (usable part) = **N3**.
- **Hammering** = N2, always. Record `hit_count`.
- **Tearing a melted bridge** = N2 (`force_method = tear`). Record whether it leaves a spike or torn edge in `tear_flag`.
- **"Easily"** means N3 with a defined handling test: lift the piece off the bench, or let it drop from the clamp after release, within **5 s**. Anything needing more is N2.

- If the piece stays on the clamp or on the kerf after unclamping, it is **not** N3.

Fields: `through_code`, `force_method` (`none` / `hand` / `tap` / `hammer` / `tear`), `hit_count`, `bridge_type` (`none` / `melt` / `slag` / `mixed`), `tear_flag` (0/1).

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6. Edge quality (apply to N2 and N3 only)

Edge quality is **several separate measurements**, not one score. ISO 9013 quality grades (thermal-cut surfaces, roughness Rz5 and perpendicularity) are the reference for comparing with industry. Verify the grade table against the standard before quoting it.

6.1 Zones

Grade every quality parameter in three zones down the thickness:

- **top** — entry, upper third
- **mid** — middle third
- **bot** — run-out, lower third

Striation commonly starts fine at the top and coarsens toward the bottom. A single average hides that.

6.2 Striation

Visual, under **10×** magnification with raking light.

Grade	Description
`S0`	None visible, even in raking light.
`S1`	Fine lines visible only in raking light.
`S2`	Lines visible in normal light.
`S3`	Deep, coarse striation, clearly visible.

Also record, for `S2/S3` only, **striation pitch** (mm, perpendicular to beam, mean of ≥ 5 lines) and **depth** (`rz_<zone>_um`, measured on the edge profile or by comparator gauge where a profilometer is not available).

6.3 Mirror

Mirror = $S0$ in **all three zones and** $rz \leq \mathrm{TBD}$ in all zones. `TBD` is set from the calibration set (§11) — do not fill it in from a published number.

`mirror_flag` = 1 only if both conditions hold. `mirror_flag` = 0 otherwise.

6.4 Dross

Dross is attached material on the **bottom (exit) edge**.

Grade	Meaning
`D0`	None.
`D1`	Loose — brushes or drops off with a fingernail.
`D2`	Adherent — needs a tool (scraper).
`D3`	Hard — cannot be removed without damage.

Record `dross_grade` (worst in the window) and `dross_height_mm` (optional, measured).

6.5 Oxide colour (heat tint)

`bright` / `straw` / `blue-brown` / `black`. Record once per sample on the **kept** face.

6.6 Taper and burr

- `taper_deg` — kerf angle from the perpendicular, measured on the kept face (°).
- `burr` — 0 none, 1 present (note position in `notes`).

6.7 Perpendicularity and dimensional check (optional, when measured)

- `perp_deg` — angularity of the cut face to the surface.
- `kerf_w_mm` — kerf width, if measurable.

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7. Face, side and direction

- **`face`** — which profile face of the beam is cut: `web` (`v/h` in DSTV terms), `flange` (`o/u`), or `wall` (tube/SHS side). Follow `docs/canonical-orientation.md` §3 for the map to DSTV letters.
- **`kept` / `scrap`** — the face is on the piece we keep, or on the piece we throw away. The two faces of one kerf differ, so log which one is graded.
- **`direction`** — travel direction on the machine: `+x`, `-x`, `+y`, `-y` (or `CW` / `CCW` for curves). Always log, never infer.

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8. Curves and corners

8.1 Curves

- `curve_side` — `inside` or `outside` of the curve, relative to the **kept** piece.
- `curve_radius_mm` — radius of the curve at the sample window.
- Curves are a **separate class** (`feature_class = curve`). Grade them with the same rubric, compared with a straight baseline cut in the same series.

8.2 Corners

Corners are their own class (`feature_class = corner`). Never average them into straight data.

- `corner_type` — `convex` (outside corner of the kept piece) or `concave` (inside corner).
- `corner_angle_deg` — 90 for most sections; record actual.
- `corner_radius_mm` — 0 if sharp.
- `approach_v_m_min` — speed before the corner.
- `dwell_ms` — any programmed dwell or power ramp at the corner (0 if none).
- Grade within **±5 mm** of the corner vertex and compare with the straight baseline cut in the same series.

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9. Feature class (required on every row)

`feature_class` ∈ `straight` · `curve` · `corner` · `hole` · `end` (end-cut or run-out only). Only `straight` rows feed the first map set. Others feed later maps (§12).

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10. Series header

Series fields are **repeated on every row** of `runs.csv` (flat CSV, no header blocks), so each row is self-describing. Fields: `series_id`, `machine`, `optics_id`, `nozzle_id`, `nozzle_dn_mm`, `gas_grade`, `gas_purity_pct`, `material_grade`, `section`, `t_nom_mm`, `feature`, `date`, `operator`. Fill `TBD` where a value is not yet known; do not guess.

Column order is fixed; the full header is the first line of `data/cut-regime/runs.csv`. Blank = not measured or not applicable. Never put commas inside a field (use `;`).

10a. Pilot series `S01_150UB14_web` (planned, not yet run)

Nine planned rows in `runs.csv` (R01–R09): five speeds at 3000 W (S sweep, R01/R03/R05/R06/R07), two power checks at v = 2.8 m/min (R02 at 2400 W, R04 at 3600 W), and three runs at the centre point (R06, R08, R09: 25%, 3000 W, 2.8 m/min). R06 is the baseline; R08 and R09 are repeats for run-to-run and grader repeatability. Planned S_beam values are in each row's `notes`. Run in a random order (not R01 to R09) and record actual execution order in `date/notes`.

Before running: confirm the % → W mapping (the pilot assumes 25% = 3000 W) and fill the TBD fields.

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11. Calibration set (to set `TBD` thresholds)

Before the first map:

1. Cut 5 samples spanning from clearly striated to clearly mirror, on one series.
2. Grade them blind (another person scores `S` grades and `through_code`).
3. Measure `rz` on each zone with the same instrument.
4. Set the mirror `rz` threshold at the value that best separates `S0` from `S1/S2`. Record the chosen value and the date in §12.

If two graders disagree on `S` grade by more than one step, retrain on the reference set before collecting.

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12. Change log

Date	Version	Change
2026-11-09	v0.1	First draft. Outcome codes N0–N3, striation S0–S3, dross D0–D3, zones, curve/corner classes. Mirror `rz` threshold TBD.
2026-11-09	v0.2	Pilot centre-point count corrected (three runs). Paper sheet moved to `cut-regime-data-sheets.md`.
2026-11-09	v0.4	Focus sign set to physical (negative = below top surface, shop practice). `cam_focus_mm` column added to `runs.csv`.
2026-11-09	v0.3	Gas pressure gauge/absolute stated. Sheet 0 gains gas grade, pack ID and pack pressure, and a focus-sign check.

Appendix A — Paper sheets

Moved to `docs/cut-regime-data-sheets.md` (Sheets 0–4). That file is the only printable form, so the spec and the sheets cannot drift apart.